Neel Tamtam

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github | linkedin | website

Skills & Certifications

Certifications CUDA Accelerated Computing | JLPT N1
Languages C | Rust | C++ | CUDA | Zig | ARM Assembly | RISC-V Assembly | Python
Embedded FreeRTOS | STM32 | ESP32
Graphics Vulkan | OpenGL | GLSL | HLSL
Cloud AWS | GCP | Docker | Apache Airflow
Backend PostgreSQL | MongoDB | GraphQL | TCP/IP | WebSockets | REST
Frameworks Axum | Leptos | PyTorch | TensorFlow

Experience

National Institutes of Health — **July 2024 - Present**
Software Engineer, National Library of Medicine — *Remote*

- Accelerated ML training pipeline with CUDA-enabled PyTorch, cutting latency 80% across 4.5M device records.
- Migrated batch processes to Apache Airflow, reducing average job runtime by 40% via parallel task execution.
- Spearheaded anomaly detection from development to board presentation, securing $1M+ in NIH funding.
- Provisioned AWS/GCP infrastructure for scalable processing across millions of FDA-regulated drugs and devices.

Notable Projects

CUDA-Accelerated Ray Tracing Simulator — **GitHub**
Graphics Programming, GPU Computing — *C++20, CUDA, Vulkan, ImGui*

- Architected a real-time ray-tracing engine using CUDA compute alongside a Vulkan backend for high efficiency.
- Optimized CUDA kernels for complex lighting, minimizing divergence and enforcing memory coalescing.
- Accelerated intersection queries with a GPU-optimized BVH, enabling real-time rendering of complex meshes.
- Engineered zero-copy Vulkan-CUDA interop using external memory sharing for low-latency frame delivery.

Custom Real-Time Operating System — **GitHub**
Embedded Systems (STM32F407) — *C, Rust, ARM Assembly, GDB, Unity*

- Designed preemptive kernel with priority round-robin scheduling, validated by a Rust FFI traffic simulator.
- Achieved deterministic task execution with <50µs scheduling jitter and 100% compliance for critical tasks.
- Built inter-task sync primitives with blocking/non-blocking semantics for a custom CS43L22 I2C driver.
- Implemented ARM assembly context switching, preserving CPU state for up to 8 concurrent tasks in <10µs.

Multiplayer Japanese Kanji Trainer in Rust — **Live Site | GitHub**
Full Stack — *Rust, Tokio, Axum, Leptos, PostgreSQL*

- Built and deployed a multiplayer kanji game in active beta for beginner to advanced (N1) Japanese learners.
- Engineered a scalable 100% Rust full-stack architecture using Leptos (WASM), Axum, Tokio, and PostgreSQL.
- Implemented JWT auth with Argon2 hashing, rate limiting via tower-governor, and SQLx compile-time queries.
- Orchestrated low-latency, real-time game state synchronization and interactive lobby chat via WebSockets.

Contributions

gfx-rs/wgpu — **PRs**

- Fixed incorrect atomic load/store emission in GLSL/HLSL backends using atomicOr/atomicExchange polyfills.
- Replaced explicit Vulkan swapchain destroy calls with Rust-native Drop impl for automatic RAII cleanup.

ziglang/zig — **PRs**

- Ported POSIX doubly-linked list functions insque/remque from musl/wasi-libc C to native Zig implementation.
- Added ABI register name aliases (ra, sp, a0–a7, etc.) as valid inline assembly clobbers for RISC-V targets.

Education

Bachelor of Computer Science and Japanese — **Aug 2020 - May 2024**
Austin College — *Texas, USA*

Semester Abroad - Japanese Language — **Jan 2023 - May 2023**
Osaka Daigakuin University — *Osaka, Japan*

Summer Abroad - Machine Learning — **June 2023 - Aug 2023**
Yonsei University — *Seoul, South Korea*